

From four levels to the Dicke model

the adiabatic elimination, done explicitly

September 14, 2026

Abstract

A companion note reduces the four-level cavity Rydberg-EIT problem all the way to a Dicke model. Here we stop two steps earlier, at the three-level cavity Hamiltonian $H^{(3)}$ of Eq. (26), and spend the space instead on Step 3 — the adiabatic elimination of the intermediate state $|2\rangle$ — which is where the Raman coupling $\lambda = g_0\Omega_c/2\Delta$, the light shifts, and the residual spontaneous emission all come from. Doing it explicitly pays for itself three times: the rank-one structure of the coupling shows that λ and the dominant systematic cannot be tuned independently; the same algebra returns the error of the reduction with no fitted constant, verified here against exact diagonalisation; and the invariant $\lambda^2/\gamma_3 = g_0^2/\Gamma_e$ shows that neither the detuning nor the control power improves the sensor, only atom number and cavity finesse. Throughout $\hbar = 1$.

1 The exact Hamiltonian

Four atomic levels with laboratory energies $\omega_1 = 0 < \omega_2 < \omega_3 < \omega_4$:

$$|1\rangle = |5S_{1/2}\rangle, \quad |2\rangle = |5P_{3/2}\rangle, \quad |3\rangle = |53D_{5/2}\rangle, \quad |4\rangle = |54P_{3/2}\rangle,$$

numerically $\omega_2/2\pi = 384.25$, $\omega_3/2\pi = 1008.82$ THz, and $(\omega_4 - \omega_3)/2\pi \simeq 10$ GHz. Writing $\sigma_{jk} = |j\rangle\langle k|$, and with no approximations whatsoever,

$$H = \underbrace{\sum_{j=1}^4 \omega_j \sigma_{jj}}_{\text{free}} + \underbrace{g_0 (a + a^\dagger)(\sigma_{12} + \sigma_{21})}_{\text{cavity, } |1\rangle \leftrightarrow |2\rangle} + \underbrace{\Omega_c \cos(\nu_c t) (\sigma_{23} + \sigma_{32})}_{\text{control laser}} + \underbrace{\Omega_\mu \cos(\nu_\mu t) (\sigma_{34} + \sigma_{43})}_{\text{microwave}}. \quad (1)$$

The classical fields carry their full time dependence; the cavity coupling carries both a and $a^\dagger$. Nothing has been dropped.

2 Step 1: the rotating frame

Transform with $\tilde{H} = UHU^\dagger + i\dot{U}U^\dagger$, choosing

$$U(t) = \exp\left\{it[\omega_L a^\dagger a + \omega_L \sigma_{22} + (\omega_L + \nu_c) \sigma_{33} + (\omega_L + \nu_c + \nu_\mu) \sigma_{44}]\right\}, \quad (2)$$

i.e. each level is rotated at the *sum of the drive frequencies needed to reach it* (ω_L is a reference near the cavity resonance). This is an exact change of picture. Its whole effect is on the diagonal:

level	rotated at	lab energy	rotating-frame energy
$ 1\rangle$	0	0	0
$ 2\rangle$	ω_L	$2\pi \times 384$ THz	$\Delta \equiv \omega_2 - \omega_L$ ($\sim$ GHz)
$ 3\rangle$	$\omega_L + \nu_c$	$2\pi \times 1009$ THz	$\delta \equiv \omega_3 - \omega_L - \nu_c$ ($\sim$ MHz)
$ 4\rangle$	$\omega_L + \nu_c + \nu_\mu$	$\omega_3 + 2\pi \times 10$ GHz	$\delta + \Delta_\mu, \quad \Delta_\mu \equiv (\omega_4 - \omega_3) - \nu_\mu$

This answers the obvious objection. $|3\rangle$ and $|4\rangle$ are *not* degenerate — they are 10 GHz apart. But the frame rotates $|4\rangle$ faster than $|3\rangle$ by exactly ν_μ , and ν_μ is chosen close to that splitting. The 10 GHz is subtracted off; what remains on the diagonal is only the residual *microwave detuning* Δ_μ , of order MHz. Setting $\Delta_\mu = 0$ makes the two entries equal — and that is a statement about the *drive*, not about the atom.

3 Step 2: the rotating-wave approximation

In the new frame the cavity term $a^\dagger \sigma_{21}$ picks up $e^{2i\omega_L t}$, and $\cos(\nu t) \sigma_{jk}$ splits into a static piece and one at $e^{2i\nu t}$. Dropping the fast pieces requires

$$g_0, \Omega_c, \Omega_\mu \ll \omega_L, \nu_c, \nu_\mu, \quad (\text{A1})$$

which is spectacularly well satisfied (MHz against THz, and MHz against the 10 GHz microwave). The result is time-independent and still exact to this order:

$$\tilde{H} = \omega_c a^\dagger a + \begin{pmatrix} 0 & g_0 a^\dagger & 0 & 0 \\ g_0 a & \Delta & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \delta & \Omega_\mu/2 \\ 0 & 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \omega_c \equiv \omega_{\text{cav}} - \omega_L. \quad (3)$$

Note the matrix is tridiagonal: each field connects only neighbouring levels.

4 Step 3: eliminate the intermediate state, in full

This is the step that does the real work, so we do it slowly. Nothing below is new physics; the point is that every factor in $\lambda = g_0 \Omega_c / 2 \Delta$ can be traced, and that the same algebra hands us the size of the error and the price paid in spontaneous emission.

4.1 Partition the space

Split the four levels into the ones we want to keep and the one we want gone,

$$P = \sigma_{11} + \sigma_{33} + \sigma_{44}, \quad Q = \sigma_{22}, \quad P + Q = \mathbf{1}, \quad PQ = 0. \quad (4)$$

Reordering Eq. (3) as $(|1\rangle, |3\rangle, |4\rangle; |2\rangle)$ puts it in block form

$$\tilde{H} = \omega_c a^\dagger a + \begin{pmatrix} H_P & V \\ V^\dagger & \Delta \end{pmatrix}, \quad H_P = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \delta & \Omega_\mu/2 \\ 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad V = \begin{pmatrix} g_0 a^\dagger \\ \Omega_c/2 \\ 0 \end{pmatrix}. \quad (5)$$

V is a 3×1 column whose entries are cavity operators: it is the only thing that connects the two blocks, and $|4\rangle$ does not appear in it at all — the microwave never touches $|2\rangle$. Everything that follows is built from V .

4.2 The elimination

Expand the state as $|\Psi\rangle = \sum_j |j\rangle \otimes |\phi_j\rangle$, where the $|\phi_j\rangle$ are (unnormalised) cavity states. Reading the rows of Eq. (3) off directly,

$$i \partial_t |\phi_1\rangle = \omega_c a^\dagger a |\phi_1\rangle + g_0 a^\dagger |\phi_2\rangle, \quad (6)$$

$$i \partial_t |\phi_2\rangle = \omega_c a^\dagger a |\phi_2\rangle + \Delta |\phi_2\rangle + g_0 a |\phi_1\rangle + \frac{\Omega_c}{2} |\phi_3\rangle, \quad (7)$$

$$i \partial_t |\phi_3\rangle = \omega_c a^\dagger a |\phi_3\rangle + \delta |\phi_3\rangle + \frac{\Omega_c}{2} |\phi_2\rangle + \frac{\Omega_\mu}{2} |\phi_4\rangle, \quad (8)$$

$$i \partial_t |\phi_4\rangle = \omega_c a^\dagger a |\phi_4\rangle + (\delta + \Delta_\mu) |\phi_4\rangle + \frac{\Omega_\mu}{2} |\phi_3\rangle. \quad (9)$$

Only Eq. (7) has a large coefficient. Move to the frame that removes the slow diagonal terms, $|\phi_j\rangle = e^{-i\omega_c a^\dagger a t} e^{-i\epsilon_j t} |\chi_j\rangle$ with $\epsilon_{1,3,4} = 0, \delta, \delta + \Delta_\mu$. Equation (7) becomes

$$i \partial_t |\chi_2\rangle = \Delta |\chi_2\rangle + \underbrace{g_0 a e^{-i\omega_c t} |\chi_1\rangle + \frac{\Omega_c}{2} e^{-i\delta t} |\chi_3\rangle}_{S(t)}, \quad (10)$$

a driven first-order equation whose source $S(t)$ has amplitude $\sim g_0 \sqrt{\langle n \rangle}, \Omega_c/2$ and turns over at rates ω_c, δ . If $|\Delta|$ dominates all four of those, $|\chi_2\rangle$ has no time to build up: it locks to the instantaneous value that makes the right-hand side vanish, $|\chi_2\rangle \simeq -S(t)/\Delta$. Back in the original frame,

$$\boxed{|\phi_2\rangle \simeq -\frac{1}{\Delta} \left(g_0 a |\phi_1\rangle + \frac{\Omega_c}{2} |\phi_3\rangle \right)} \quad (11)$$

$|2\rangle$ is no longer a degree of freedom; it is a *shadow* of the two states that drive it. Substitute Eq. (11) wherever $|\phi_2\rangle$ appears. In Eq. (6),

$$g_0 a^\dagger |\phi_2\rangle = -\frac{g_0^2}{\Delta} a^\dagger a |\phi_1\rangle - \frac{g_0 \Omega_c}{2\Delta} a^\dagger |\phi_3\rangle, \quad (12)$$

and in Eq. (8),

$$\frac{\Omega_c}{2} |\phi_2\rangle = -\frac{g_0 \Omega_c}{2\Delta} a |\phi_1\rangle - \frac{\Omega_c^2}{4\Delta} |\phi_3\rangle. \quad (13)$$

Two terms of each kind, four in all, and they assemble into a 3×3 matrix in $\{|1\rangle, |3\rangle, |4\rangle\}$:

$$\tilde{H}^{(3)} = \omega_c a^\dagger a + \begin{pmatrix} -\frac{g_0^2 a^\dagger a}{\Delta} & -\lambda a^\dagger & 0 \\ -\lambda a & \delta - \frac{\Omega_c^2}{4\Delta} & \Omega_\mu/2 \\ 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \boxed{\lambda = \frac{g_0 \Omega_c}{2\Delta}} \quad (14)$$

Note that the $|1\rangle$ shift came out *normally ordered*, $a^\dagger a$ and not $aa^\dagger$: the photon is absorbed before it is re-emitted, so an empty cavity is not shifted. Had we eliminated $|2\rangle$ carelessly we might have written $n+1$ where the correct answer is n .

4.3 The same result in one line

Equation (14) is the second-order term of the Feshbach (Schur-complement) reduction of the block matrix (5),

$$H_{\text{eff}}(E) = H_P + V (E - \Delta)^{-1} V^\dagger \xrightarrow{E \rightarrow 0} H_P - \frac{V V^\dagger}{\Delta}, \quad (15)$$

and the outer product does the bookkeeping for us in one step:

$$V V^\dagger = \begin{pmatrix} g_0 a^\dagger \\ \Omega_c/2 \\ 0 \end{pmatrix} (g_0 a \quad \Omega_c/2 \quad 0) = \begin{pmatrix} g_0^2 a^\dagger a & \frac{g_0 \Omega_c}{2} a^\dagger & 0 \\ \frac{g_0 \Omega_c}{2} a & \Omega_c^2/4 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (16)$$

Divide by $-\Delta$ and add H_P : Eq. (14), identically. Rank one, which is why the Raman coupling is not independent of the two light shifts — a fact we return to below.

4.4 Reading the three new terms

$-\lambda(a^\dagger\sigma_{13} + a\sigma_{31})$, **the Raman coupling**. A cavity photon is absorbed on $|1\rangle \rightarrow |2\rangle$ and a control photon is emitted on $|2\rangle \rightarrow |3\rangle$, with $|2\rangle$ visited only virtually. The product of the two matrix elements divided by the energy denominator is $g_0 \cdot (\Omega_c/2)/\Delta$. This is the term the whole scheme is built on, and it is *first* order in each field: halving the control power halves λ . The overall minus sign is a phase convention ($|3\rangle \rightarrow -|3\rangle$ removes it) and we drop it from here on.

$-\Omega_c^2/4\Delta$ on $|3\rangle$, **the control light shift**. Absorbed by redefining the two-photon detuning, $\delta \rightarrow \delta - \Omega_c^2/4\Delta$. Harmless as bookkeeping, not harmless as physics — see the box at the end of this section.

$-(g_0^2/\Delta)a^\dagger a$ on $|1\rangle$, **the dispersive shift**. This one multiplies $a^\dagger a$, so it is not an atomic energy at all but a shift of the *cavity*. With N atoms sitting in $|1\rangle$ it adds up:

$$\omega_c \longrightarrow \omega_c - \frac{Ng_0^2}{\Delta} = \omega_c \left(1 - \frac{Ng_0^2}{\Delta\omega_c}\right). \quad (17)$$

For $N = 10^6$, $g_0/2\pi = 1$ MHz and $\Delta/2\pi = 100$ MHz this is $2\pi \times 10$ GHz, vastly larger than $\omega_c/2\pi \sim 1$ MHz. It is not a perturbation to be neglected; it is a redefinition of where the cavity sits, and in the laboratory it is absorbed by locking the cavity to the shifted resonance. What matters for the sensor is only that it drifts with N and with Δ .

4.5 Keeping the loss

State $|2\rangle$ is not merely inconvenient, it radiates at $\Gamma_e/2\pi \simeq 6$ MHz. Nothing above changes if we track that by letting the denominator go complex, $\Delta \rightarrow \Delta - i\Gamma_e/2$, which is the standard non-Hermitian bookkeeping for a level whose population leaves the system. Expanding for $|\Delta| \gg \Gamma_e$,

$$\frac{1}{\Delta - i\Gamma_e/2} \simeq \frac{1}{\Delta} + \frac{i\Gamma_e}{2\Delta^2}, \quad (18)$$

so each entry of Eq. (16) acquires an imaginary part. Two of them are the price of the elimination:

$$\boxed{\gamma_3 = \frac{\Omega_c^2}{4\Delta^2} \Gamma_e \quad (\text{Rydberg state}), \quad \kappa_{\text{abs}} = \frac{g_0^2}{\Delta^2} \Gamma_e \quad (\text{cavity})} \quad (19)$$

Both are Γ_e multiplied by the residual $|2\rangle$ population that the corresponding drive maintains, $(\Omega_c/2\Delta)^2$ and $(g_0/\Delta)^2$ per intracavity photon respectively; the elimination has not removed the decay, only diluted it. At $\Omega_c/2\pi = 5$ MHz, $g_0/2\pi = 1$ MHz, $\Delta/2\pi = 100$ MHz these are $2\pi \times 3.75$ and 0.60 kHz, confirmed against the exact eigenvalues to better than 0.5%.

The scaling looks at first like a free lunch — $\lambda \propto \Delta^{-1}$ but $\gamma_3 \propto \Delta^{-2}$, so detuning further seems to buy coherence faster than it costs coupling. It does not. Form the combination in which the Δ 's and Ω_c 's cancel:

$$\boxed{\frac{\lambda^2}{\gamma_3} = \frac{g_0^2 \Omega_c^2 / 4\Delta^2}{\Omega_c^2 \Gamma_e / 4\Delta^2} = \frac{g_0^2}{\Gamma_e}} \quad (20)$$

independent of *both* knobs the elimination gives us. Detuning further and raising the control power to compensate slides λ and γ_3 along a line of constant λ^2/γ_3 ; Step 3 provides no leverage on it whatsoever. Divide by the cavity linewidth and, with the collective enhancement $\lambda \rightarrow \lambda\sqrt{N}$ at fixed single-atom γ_3 , this is the familiar figure of merit and the only one in the problem:

$$\frac{N\lambda^2}{\kappa\gamma_3} = \frac{Ng_0^2}{\kappa\Gamma_e} = NC. \quad (21)$$

Atom number and cavity finesse are what improve the sensor; Δ and Ω_c only decide where on the line one sits. The cavity absorption is the exception that proves the point — $\lambda^2/\kappa_{\text{abs}} = \Omega_c^2/4\Gamma_e$ *does* reward control power, because κ_{abs} has no Ω_c in it.

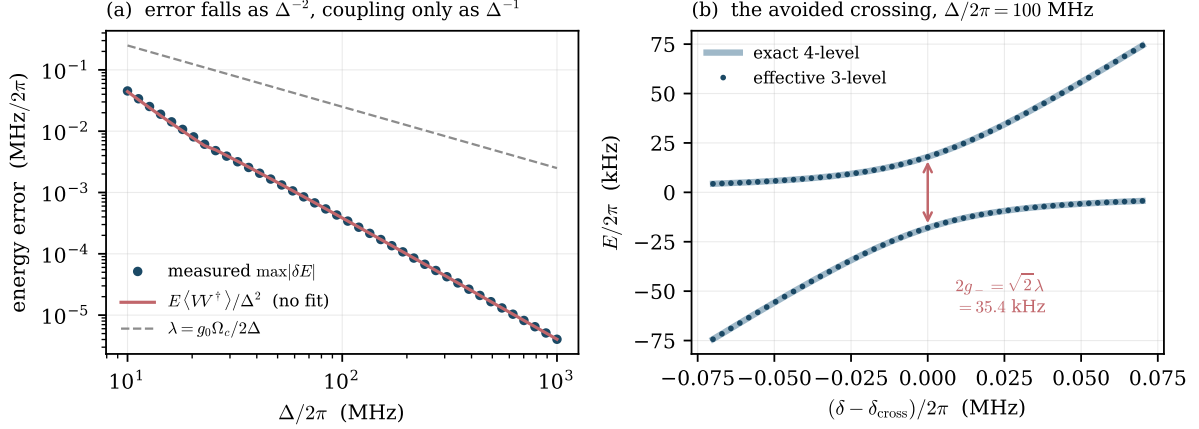


Figure 1: **Step 3 tested against exact diagonalisation**, in the single-excitation sector at $g_0/2\pi = 1$, $\Omega_c/2\pi = 5$, $\Omega_\mu/2\pi = 2$, $\delta/2\pi = 0.3$ MHz, $\Delta_\mu = 0$. (a) Worst-case eigenvalue error versus Δ (points) against Eq. (22) with no fitted parameter (line); the Raman coupling λ is shown for scale, falling only as Δ^{-1} . (b) The avoided crossing at $\Delta/2\pi = 100$ MHz. Measured minimum gap 35.9 kHz against the predicted $2g_- = \sqrt{2}\lambda = 35.4$ kHz; the 1.5% excess is the $\Omega_c^2/4\Delta$ light shift, which sits on $|3\rangle$ alone and so slightly reshapes the dressing of Step 4.

4.6 How good is it?

The one thing dropped between Eqs. (15) and (14) is the energy dependence of the resolvent. Restoring it,

$$(E - \Delta)^{-1} = -\frac{1}{\Delta} \left(1 + \frac{E}{\Delta} + \dots \right) \implies \boxed{\delta E \simeq \frac{E \langle VV^\dagger \rangle}{\Delta^2}, \quad \langle VV^\dagger \rangle = g_0^2 \langle a^\dagger a \rangle P_1 + \frac{\Omega_c^2}{4} P_3} \quad (22)$$

with P_1, P_3 the populations of $|1\rangle, |3\rangle$ in the state whose energy is E . Two things are worth noticing. The error is second order in $1/\Delta$ while λ is only first order, so the model gets *relatively* better as Δ grows even though the physics it describes gets weaker. And a state with no $|3\rangle$ component — a photon with the atom in $|1\rangle$ — is described far more accurately than the dressed states, since it is only g_0 , not Ω_c , that reaches $|2\rangle$.

Figure 1 tests both statements against exact diagonalisation. The rotating-wave Hamiltonian (3) conserves $N_{\text{exc}} = a^\dagger a + \sigma_{22} + \sigma_{33} + \sigma_{44}$, so its single-excitation sector ($|1, n=1\rangle, |2, 0\rangle, |3, 0\rangle, |4, 0\rangle$) is a closed 4×4 problem and the comparison needs no truncation. Panel (a): Eq. (22) predicts the error with no adjustable constant, over two decades. Panel (b): the avoided crossing between the cavity and the lower dressed branch, whose width is the quantity Step 3 exists to produce.

4.7 When is it allowed?

Collecting the conditions used above,

$$|\Delta| \gg \Omega_c, \quad g_0 \sqrt{\langle a^\dagger a \rangle}, \quad \Omega_\mu, \quad |\delta|, \quad |\omega_c|, \quad \Gamma_e. \quad (\text{A2})$$

The first two are the amplitudes of the source $S(t)$, the next three its turnover rates, and the last is the width of the state being eliminated. It is the *full* list: a treatment that checks only $|\Delta| \gg \Omega_c, \Gamma_e$ will be caught out at high intracavity photon number, where $g_0 \sqrt{\langle n \rangle}$ creeps up on Δ . With $\Delta/2\pi = 100$ MHz and $g_0/2\pi = 1$ MHz the elimination survives to $\langle n \rangle \sim 10^3$ and no further.

Where the dominant systematic comes from. $VV^\dagger$ in Eq. (16) has rank one, so the Raman coupling and the light shift are not independent knobs: $\lambda \propto \Omega_c$ and the shift $\propto \Omega_c^2$ both track the control-laser intensity, and the shift adds *directly* to δ — the same channel the measurand uses. Since $\Omega_c^2 \propto I$, a fractional intensity fluctuation ε moves δ by $\varepsilon \Omega_c^2 / 4\Delta$, at our numbers $\varepsilon \times 2\pi \times 62$ kHz. Control-laser intensity noise is therefore structurally indistinguishable from signal, and no amount of care with the microwave will fix it.

5 Step 4: dress the Rydberg pair

The lower-right block of Eq. (14) is a two-level problem on its own:

$$H_\mu = \begin{pmatrix} \delta & \Omega_\mu/2 \\ \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix} = \left(\delta + \frac{\Delta_\mu}{2}\right) \mathbf{1} + \frac{1}{2}(\Omega_\mu \sigma_x + \Delta_\mu \sigma_z). \quad (23)$$

Diagonalise with the rotation $R(\theta)$ defined by

$$\tan \theta = \frac{\Omega_\mu}{\Delta_\mu}, \quad |+\rangle = \sin \frac{\theta}{2} |3\rangle + \cos \frac{\theta}{2} |4\rangle, \quad |-\rangle = \cos \frac{\theta}{2} |3\rangle - \sin \frac{\theta}{2} |4\rangle, \quad (24)$$

giving the *generalised Rabi* splitting

$$\omega_\pm = \delta + \frac{\Delta_\mu}{2} \pm \frac{W}{2}, \quad W = \sqrt{\Omega_\mu^2 + \Delta_\mu^2} \quad (25)$$

Now rotate the rest of Eq. (14). The Raman coupling connects $|1\rangle$ to the *bare* $|3\rangle = \sin \frac{\theta}{2} |+\rangle + \cos \frac{\theta}{2} |-\rangle$, so it splits between the branches in that ratio. In the basis $(|g\rangle, |+\rangle, |-\rangle)$:

$$H^{(3)} = \begin{pmatrix} 0 & g_+ a^\dagger & g_- a^\dagger \\ g_+ a & \omega_+ & 0 \\ g_- a & 0 & \omega_- \end{pmatrix}, \quad g_+ = \lambda \sin \frac{\theta}{2}, \quad g_- = \lambda \cos \frac{\theta}{2} \quad (26)$$

There is no $|+\rangle \leftrightarrow |-\rangle$ entry: that transition is the microwave one and carries no cavity matrix element.

Equation (26) is where this note stops. It is still excitation-number conserving, so on its own it radiates nothing; turning it into a Dicke model requires counter-rotating terms $a^\dagger b^\dagger + ab$ from a second drive tone, and that, together with the Holstein–Primakoff step and the superradiant threshold, is the subject of the companion note.